Plan

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Row Echelon Form: A matrix A is said to be in row echelon form if it satisfies the following properties:

1. All rows consisting entirely of 0's are at the bottom.

2. In each non-zero row, the first non-zero entry (called the leading entry or pivot) is in a column to the left (**strictly**) of any leading entry below it. [The columns containing a leading entry is called a leading column].

- The following matrices are in row echelon form:

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 3 & 2 \\ 0 & 0 & 5 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- If a matrix A is in row echelon form, then in each column of A containing a leading entry, the entries below that leading entry are zero.

- The following matrices are **not** in row echelon form:

$$\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 & -1 & 2 \\ 1 & 0 & 5 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Elementary Row Operations: The following row operations are called elementary row operation of a matrix:

1. Interchange of two rows R_i and R_j (notation $R_i \leftrightarrow R_j$).
2. Multiply a row R_i by a non-zero constant c ($R_i \to cR_i$).
3. Add a multiple of a row R_j to another row R_i ($R_i \to R_i + cR_j$).

Example

Transform the following matrix to row echelon form

$$\begin{bmatrix} 0 & 2 & 3 & 8 \\ 2 & 3 & 1 & 5 \\ 1 & -1 & -2 & -5 \end{bmatrix}.$$

- A given matrix can be reduced to several row echelon forms.

Row Equivalent Matrices: Matrices A and B are said to be row equivalent if there is a finite sequence of elementary row operations that converts A into B or B into A.

Result

*Matrices A and B are row equivalent if and only if (**iff**) they can be reduced to the same row echelon form.*

Linear System of Equations:

A linear system of m equations in n unknowns $x_1, x_2, \ldots, x_n$ is a set of equations of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots \qquad\qquad \vdots \qquad\qquad\quad \vdots \quad &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m, \end{aligned} \tag{1}$$

where $a_{ij}, b_i \in \mathbb{C}$ for each $1 \leq i \leq m$ and $1 \leq j \leq n$. Letting

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix},$$

we can represent the above system of equations as $A\mathbf{x} = \mathbf{b}$.

- The matrix A is called the coefficient matrix of the system of equations $A\mathbf{x} = \mathbf{b}$.

- The matrix $[A|\ \mathbf{b}]$, as given below, is called the augmented matrix of the system of equations $A\mathbf{x} = \mathbf{b}$.

$$[A|\ \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array}\right].$$

- The vertical bar is used in the augmented matrix $[A|\ \mathbf{b}]$ only to distinguish the column vector $\mathbf{b}$ from the coefficient matrix A.

- If $\mathbf{b} \neq \mathbf{0}$ then $A\mathbf{x} = \mathbf{b}$ is called a non-homogeneous system of equations.

- If $\mathbf{b} = \mathbf{0} = [0, 0, \ldots, 0]^t$, i.e., if $b_1 = b_2 = \ldots = b_m = 0$, the system $A\mathbf{x} = \mathbf{0}$ is called a homogeneous system of equations.

- A solution of the linear system $A\mathbf{x} = \mathbf{b}$ is a column vector $\mathbf{y} = [y_1, y_2, \ldots, y_n]^t$ such that the linear system (1) is satisfied by substituting y_i in place of x_i. That is, $A\mathbf{y} = \mathbf{b}$ holds true.

- The solution $\mathbf{0}$ of $A\mathbf{x} = \mathbf{0}$ is called the **trivial** solution and any other solution of $A\mathbf{x} = \mathbf{0}$ is called a **non-trivial** solution.
- Two system of linear equations are called equivalent if their augmented matrices are row-equivalent.
- An elementary row operation on $A\mathbf{x} = \mathbf{b}$ is an elementary row operation on the augmented matrix $[A \mid \mathbf{b}]$.

Result

Let $C\mathbf{x} = d$ be the linear system obtained from the linear system $A\mathbf{x} = b$ by a single elementary row operation. Then the linear systems $A\mathbf{x} = b$ and $C\mathbf{x} = d$ have the same set of solutions.

Result

Two equivalent system of linear equations have the same set of solutions.

Gaussian Elimination Method: Use the following steps to solve a system of equations $A\mathbf{x} = \mathbf{b}$.

1. Write the augmented matrix $[A \mid \mathbf{b}]$.
2. Use elementary row operations to reduce $[A \mid \mathbf{b}]$ to row echelon form.
3. Use back substitution to solve the equivalent system that corresponds to the row echelon form.

Example

Use Gaussian Elimination method to solve the system:

$(a)\quad y + z = 1, \quad x + y + z = 2, \quad x + 2y + 2z = 3$

$(b)\quad y + z = 1, \quad x + y + z = 2, \quad x + 2y + 3z = 4$

$(c)\quad y + z = 1, \quad x + y + z = 2, \quad x + 2y + 2z = 4.$

- If $A\mathbf{x} = \mathbf{b}$ has at least one solution then it is called a consistent system. Else, it is called an inconsistent system.

Reduced Row Echelon Form: A matrix A is said to be in reduced row echelon form(RREF) if it satisfies the following:

1. A is in row echelon form.
2. The leading entry in each non-zero row is a 1.
3. Each column containing a leading 1 has zeros elsewhere.

- The following matrices are in reduced row echelon form

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Leading and Free Variable:

- Consider the system $A\mathbf{x} = \mathbf{b}$ in n variables and m equations.
- Let $[R \mid \mathbf{r}]$ be a row echelon form of $[A \mid \mathbf{b}]$.
- The variables corresponding to the leading columns in the first n columns of $[R \mid \mathbf{r}]$ are called the leading variables or basic variables. Remaining variables are called free variables.

- If $A = [B \mid C]$, and $RREF(A) = [R_1 \mid R_2]$, then note that R_1 is the RREF of B.

Result

Every matrix has a unique reduced row echelon form.

Method of Induction: [Version I] (Recall)

- Let $P(n)$ be a mathematical statement based on all positive integers n.
- Suppose that $P(1)$ is true.
- For $k \geq 2$, suppose $P(k-1)$ is true implies that $P(k)$ is also true.

Then the statement $P(n)$ is true for all positive integers.

Proof: RREF is unique (Using Induction)

We will apply induction on number of columns.

Base Case: Consider a matrix with one column i.e. $A_{m\times 1}$. Observe that its RREF is unique. What are the possibilities? $\mathbf{O}_{m\times 1}$ and e_1. Can you transform $\mathbf{O}_{m\times 1}$ into e_1?

Induction Hypothesis: Let any matrix having $k-1$ columns have a unique RREF.

Inductive Case:

- Consider a matrix $A_{m\times k}$.
- We need to show that A also has unique RREF.
- Assume A has two RREF say B and C. Note that because of induction hypothesis the submatrices B_1 and C_1 formed by the first $k-1$ columns of B and C, respectively, are identical. In particular, they have same number of nonzero rows.

- We have that the systems $A\mathbf{x} = 0$, $B\mathbf{x} = 0$ and $C\mathbf{x} = 0$ have the same set of solutions. Let the i^{th} row of the k^{th} column be different in B and C i.e. $b_{ik} \neq c_{ik} \Rightarrow b_{ik} - c_{ik} \neq 0$.

$$\begin{bmatrix} & & & \\ & & & b_{ik} \\ & & & \end{bmatrix} \quad \begin{bmatrix} & & & \\ & & & c_{ik} \\ & & & \end{bmatrix}$$

  Let $\mathbf{u}$ be an arbitrary solution of $A\mathbf{x} = 0$.
  $\Rightarrow B\mathbf{u} = 0$, $C\mathbf{u} = 0$ and $(B - C)\mathbf{u} = 0$
  $\Rightarrow (b_{ik} - c_{ik})u_k = 0 \Rightarrow u_k$ must be 0
  $\Rightarrow x_k$ is not a free variable
  $\Rightarrow$ there must be a leading 1 in the k^{th} column of B and C
  $\Rightarrow$ the location of 1 is different in the k^{th} column
  $\Rightarrow$ numbers of nonzero rows in B_1 and C_1 are different,
  a contradiction.

Result

Matrices A and B are row equivalent iff $RREF(A) = RREF(B)$.

Method of transforming a given matrix to reduced row echelon form: Let A be an $m \times n$ matrix. The following step by step method is used to obtain the reduced row echelon form of A.

(1) Let the i-th column be the left most non-zero column of A. Interchange rows, if necessary, to make the first entry of this column non-zero. Now use elementary row operations to make all the entries below this first entry equal to 0.

(2) Forget the first row and first i columns. Start with the lower $(m-1) \times (n-i)$ sub matrix of the matrix obtained in the first step and proceed as in **Step 1**.

(3) Repeat the above steps until we get a row echelon form.

(4) Now use the leading term in each of the leading column to make (by elementary row operations) all other entries in that column equal to zero. Use this step starting form the rightmost leading column.

(5) Scale the leading entries of the matrix obtained in the previous step, by multiplying the rows by suitable constants, to make all the leading entries equal to 1, ending with the unique reduced row echelon form of A.

Gauss-Jordan Elimination Method: Use the following steps to solve a system of equations $A\mathbf{x} = \mathbf{b}$.

1. Write the augmented matrix $[A \mid \mathbf{b}]$.

2. Use elementary row operations to transform $[A \mid \mathbf{b}]$ to reduced row echelon form.

3. Use back substitution to solve the equivalent system that corresponds to the reduced row echelon form.

Example

Solve the system
$w - x - y + 2z = 1,\ 2w - 2x - y + 3z = 3,\ -w + x - y = -3$
using Gauss-Jordan elimination method.

Example

Solve the following systems using Gauss-Jordan elimination method:

(a) $2y + 3z = 8, \quad 2x + 3y + z = 5, \quad x - y - 2z = -5$

(b) $x - y + 2z = 3, \quad x + 2y - z = -3, \quad 2y - 2z = 1.$

Rank: The rank of a matrix A, denoted $\text{rank}(A)$, is the number of non-zero rows in its row echelon form.

Result

Let $A\mathbf{x} = \mathbf{b}$ be a system of equations with n variables. Then number of free variables is equal to $n - rank(A)$.

Result

Let $A\mathbf{x} = \mathbf{0}$ be a homogeneous system of equations with n variables. If $rank(A) < n$ then the system has infinitely many solutions.

Result

Let $A\mathbf{x} = \mathbf{b}$ be a system of equations with n variables.

1. *If $rank(A) \neq rank([A \mid \mathbf{b}])$ then the system $A\mathbf{x} = \mathbf{b}$ is inconsistent;*
2. *If $rank(A) = rank([A \mid \mathbf{b}]) = n$ then the system $A\mathbf{x} = \mathbf{b}$ has a unique solution; and*
3. *If $rank(A) = rank([A \mid \mathbf{b}]) < n$ then the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.*